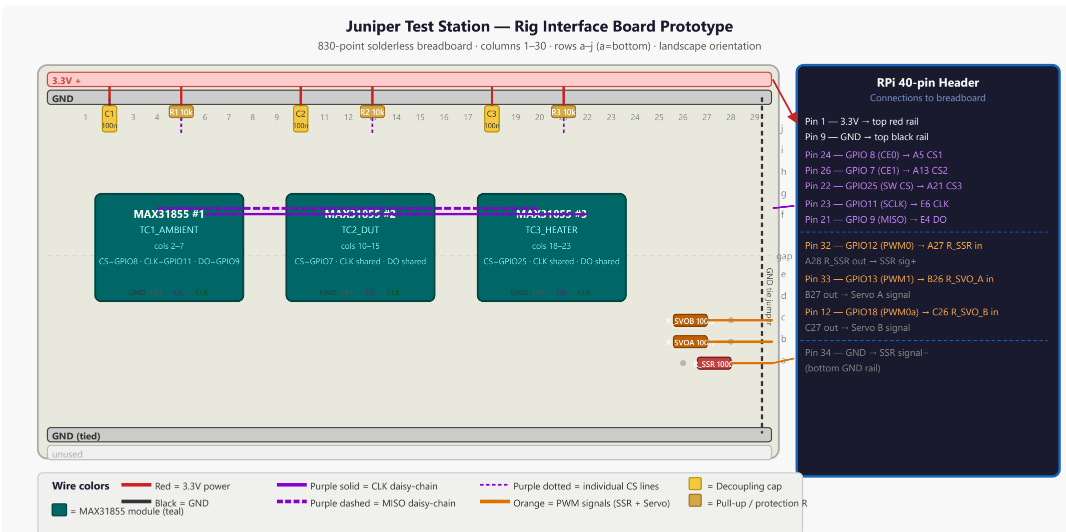
Breadboard Prototype — Rig Interface Board

> 📄 **Print-ready PDF:** `docs/pdf/breadboard-prototype.pdf`



The diagram above shows every component and wire at exact column/row coordinates. The text below covers placement, external connections, build order, and verification.

Wire Color Convention

- **Red** — 3.3V power positive (MAX31855 VCC)
- **Black** — GND (all rails)
- **Orange** — PWM signal lines (heater SSR control, servo PWM)
- **Green** — digital output lines (not used on this breadboard)
- **Blue** — digital input lines (not used on this breadboard)
- **Purple** — SPI signals (MOSI, MISO, SCLK, and all CS lines)

Coordinate System

Held in landscape orientation, long axis left-to-right:

- **Rows** are labeled `a` through `j` along the short edge — `a` **at the bottom**, `j` at the top. The center gap separates rows `e` (bottom half) and `f` (top half).
- **Columns** are numbered `1` through `30` along the long edge — **column `1` on the left**, column 30 on the right.
- A tie point is identified by `<row letter><column number>` — e.g., `E5` is row `e`, column 5 (lower area, left-center).
- Power rails: **top red rail** = 3.3V (outermost above row `j`); **top black rail** = GND (just below top red); **bottom red rail** = unused; **bottom black rail** = GND (tied to top GND rail at column 30).

This breadboard uses a standard 830-point full-size layout (63 column × a–j rows plus two split power rail pairs at top and bottom).

Component Placement

Component	Breadboard location	Notes
MAX31855 #1 (TC1_AMBIENT)	Module straddles gap, pins in rows `e` / `f`, columns 2–7	8-pin SIP: VCC=E2, GND=E3, DO=E4, CS=E5, CLK=E6. Top row (f) mirrors
MAX31855 #2 (TC2_DUT)	Module straddles gap, pins in rows `e` / `f`, columns 10–15	VCC=E10, GND=E11, DO=E12, CS=E13, CLK=E14
MAX31855 #3 (TC3_HEATER)	Module straddles gap, pins in rows `e` / `f`, columns 18–23	VCC=E18, GND=E19, DO=E20, CS=E21, CLK=E22
C1 (100nF ceramic)	Vertical: top leg **E2**, bottom leg drops to top GND rail	Decoupling on TC1 VCC — non-polarised, either orientation
C2 (100nF ceramic)	Vertical: top leg **E10**, bottom leg drops to top GND rail	Decoupling on TC2 VCC
C3 (100nF ceramic)	Vertical: top leg **E18**, bottom leg drops to top GND rail	Decoupling on TC3 VCC
R1 (10kΩ CS pull-up, TC1)	Horizontal: **A5** ↔ top red (3.3V) rail at column 5	Keeps CS1 high at startup; one leg in A5, other to top + rail
R2 (10kΩ CS pull-up, TC2)	Horizontal: **A13** ↔ top red rail at column 13	CS2 pull-up
R3 (10kΩ CS pull-up, TC3)	Horizontal: **A21** ↔ top red rail at column 21	CS3 pull-up
R_SSR (100Ω, ¼W)	Horizontal: **A27** ↔ **A28**	GPIO12 → SSR signal+ series resistor; left leg is input, right leg exits east
R_SVO_A (100Ω, ¼W)	Horizontal: **A26** ↔ **A27** (shares A27 column bus with R_SSR at row `b`) — place at **B26** ↔ **B27**	GPIO13 → Servo A PWM series resistor
R_SVO_B (100Ω, ¼W)	Horizontal: **C26** ↔ **C27**	GPIO18 → Servo B PWM series resistor

> **Module pin-out note:** Standard MAX31855 breakout boards (Adafruit or clones) expose 5 pins: VCC, GND, DO (MISO), CS, CLK — in that order from left to right. Confirm your specific breakout's silkscreen before inserting. The module body straddles the center gap; pins on the bottom-half rows (`e`) are the ones that matter for wiring since they share column buses with rows `a` – `d` .

External Connections

All wires run from the breadboard to the RPi 40-pin header (or to the off-board SSR/servo terminals) via color-coded Dupont jumpers.

Power and Ground

Breadboard tap	Wire color	Destination	Notes
Top red rail	**Red**	RPi pin 1 (3.3V)	Powers all three MAX31855 VCC pins + CS pull-ups
Top black rail	**Black**	RPi pin 9 (GND)	Common GND for all modules
Top black rail (tie)	**Black**	Bottom black rail at column 30	Single jumper to extend GND across board

SPI Bus (shared by all three MAX31855 modules)

Breadboard tap	Wire color	RPi pin	BCM GPIO	Notes
E6 (CLK, TC1) + jumpers to E14 and E22	**Purple**	Pin 23	GPIO11 (SCLK)	One wire from RPi to E6; short purple jumpers E6→E14 and E14→E22 to daisy-chain CLK
E4 (DO, TC1) + jumpers to E12 and E20	**Purple**	Pin 21	GPIO9 (MISO)	Same daisy-chain pattern for MISO

Chip-Select Lines (individual per module)

Breadboard tap	Wire color	RPi pin	BCM GPIO	Notes
A5 (CS, TC1_AMBIENT)	**Purple**	Pin 24	GPIO8 (CE0)	Hardware SPI CS0
A13 (CS, TC2_DUT)	**Purple**	Pin 26	GPIO7 (CE1)	Hardware SPI CS1
A21 (CS, TC3_HEATER)	**Purple**	Pin 22	GPIO25	Software GPIO CS — configure as output, pull high

Heater SSR and Servo Outputs

Breadboard tap	Wire color	RPi pin	BCM GPIO	Notes
A27 (R_SSR input)	**Orange**	Pin 32	GPIO12 (PWM0)	GPIO12 → R_SSR (100Ω) → SSR signal+
A28 (R_SSR output)	**Orange**	SSR terminal 3 (signal+)	—	Exits east from breadboard to SSR
B26 (R_SVO_A input)	**Orange**	Pin 33	GPIO13 (PWM1)	GPIO13 → R_SVO_A (100Ω) → Servo A signal
B27 (R_SVO_A output)	**Orange**	Servo A signal wire	—	Exits east
C26 (R_SVO_B input)	**Orange**	Pin 12	GPIO18 (PWM0 alt)	GPIO18 → R_SVO_B (100Ω) → Servo B signal
C27 (R_SVO_B output)	**Orange**	Servo B signal wire	—	Exits east
Bottom black rail (col 28)	**Black**	SSR terminal 4 (signal-)	—	SSR control GND

Build Order

Wire colors in **bold** below. Work left to right, power last-connect, power first-disconnect.

1. **Install C1, C2, C3 first** (anchor the decoupling before the ICs go in). Push each 100nF ceramic cap vertically: top leg into **E2**, **E10**, and **E18** respectively; bottom leg drops straight down to the top black (GND) rail. Non-polarised — either orientation. Verify no bridges with adjacent columns.

2. **Insert MAX31855 #1** so VCC is at **E2**, GND at **E3**, DO at **E4**, CS at **E5**, CLK at **E6**. Module body straddles the center gap. Double-check the silkscreen — DO NOT insert backwards. Confirm C1's VCC leg (E2) is on the same column bus as the module's VCC pin.

3. **Insert MAX31855 #2** at columns 10–14 (VCC=E10, GND=E11, DO=E12, CS=E13, CLK=E14). Confirm C2 at E10.

4. **Insert MAX31855 #3** at columns 18–22 (VCC=E18, GND=E19, DO=E20, CS=E21, CLK=E22). Confirm C3 at E18.

5. **Install CS pull-up resistors R1, R2, R3.** Each is 10kΩ. R1: one leg into **A5**, the other leg into the top red rail at column 5. R2: **A13** to top red rail at column 13. R3: **A21** to top red rail at column 21. These hold CS lines HIGH when the RPi is booting or the pin is not yet configured — prevents false SPI transactions.

6. **Wire the SPI bus daisy-chain.**

7. **Purple** jumper: CLK from **E6** (TC1) → **E14** (TC2). Another **purple** jumper: E14 → **E22** (TC3). CLK is now shared.

8. **Purple** jumper: DO (MISO) from **E4** → **E12**. Another **purple** jumper: E12 → **E20**. MISO is now shared.

9. **Install protection resistors R_SSR, R_SVO_A, R_SVO_B** in the columns 26–28 area.

10. R_SSR (100Ω) horizontal at **A27** ↔ **A28**.
11. R_SVO_A (100Ω) horizontal at **B26** ↔ **B27**.

12. R_SVO_B (100Ω) horizontal at **C26** ↔ **C27**.

13. **Connect power rails.** Do NOT connect RPi yet.

14. **Black** jumper: top black rail → bottom black rail at column 30 (GND tie across full board).

15. Verify no shorts across + and - rails with a DMM (continuity mode). Should read open.

16. **Connect RPi power and GND.** With the RPi powered OFF:

17. **Red** wire: RPi pin 1 (3.3V) → top red rail at column 1.

18. **Black** wire: RPi pin 9 (GND) → top black rail at column 1. Power on RPi. Measure top red rail = 3.3V, top black rail = 0V. Check E2/E10/E18 all read 3.3V.

19. **Connect all signal wires** per the External Connections tables above. RPi SPI pins last. Clip each Dupont female onto the corresponding RPi pin one at a time, confirming pin number against the 40-pin header diagram before pushing:

    - **Purple** × 3: CS1 (pin 24), CS2 (pin 26), CS3 (pin 22) → A5, A13, A21
    - **Purple** × 2: SCLK (pin 23) → E6; MISO (pin 21) → E4
    - **Orange** × 3: GPIO12 (pin 32) → A27; GPIO13 (pin 33) → B26; GPIO18 (pin 12) → C26
    - **Orange** × 2: A28 → SSR terminal 3; B27 → Servo A signal; C27 → Servo B signal
    - **Black**: bottom black rail → SSR terminal 4 (signal GND)

Verification Checklist

Stage	What to check	How to check
1 — Power rail	Top red rail = 3.3V, GND rail = 0V	DMM, probes on rail strips
2 — VCC at modules	E2, E10, E18 all = 3.3V	DMM tip on each tie point
3 — GND at modules	E3, E11, E19 all = 0V	DMM
4 — CS pull-ups	A5, A13, A21 all = 3.3V with RPi booted (before Python runs)	DMM — confirms R1/R2/R3 wired to + rail
5 — SPI CLK daisy	Probe E6, E14, E22 simultaneously while running `spidev` test — all should show same clock waveform	Oscilloscope or logic analyser; or verify each column with DMM in AC mode
6 — MISO continuity	Measure resistance E4→E12→E20 with RPi off — should be <5Ω	DMM continuity across the daisy-chain
7 — TC1 reads	`python3 -c "import board, busio, adafruit_max31855; ..."` returns ambient temp ±5°C	Software test — see `software/test_tc.py`
8 — TC2, TC3 read	Same test for GPIO7 (CS2) and GPIO25 (CS3)	Software
9 — SSR signal	RPi GPIO12 = 3.3V with `gpio write 12 1` (WiringPi) or `raspi-gpio set 12 op dh`	DMM at A28 (R_SSR output) reads 3.3V
10 — Servo signal	GPIO13 and GPIO18 toggle visible on DMM/scope at A28 column area	DMM in frequency mode at B27 / C27 output
11 — No cross-rail shorts	Power off, DMM continuity: top red rail ↔ top black rail → should be OPEN	Any continuity beep = module or component bridge — locate and fix before next power-on